

Which health benefits of the Mediterranean diet are actually supported?

ABSTRACT — Mediterranean dietary programmes have their strongest clinical evidence in cardiovascular prevention. In adults at increased risk, three endpoint trials and modern evidence syntheses indicate fewer major cardiovascular events, although absolute benefits are modest and Mediterranean programmes are not clearly superior to every well-supported low-fat programme. Type 2 diabetes is the next most coherent outcome: randomized prevention data, prospective cohorts, and objective biomarker studies point in the same direction, while established diabetes shows small improvements in glycaemia and body mass. Higher adherence is consistently associated with longer survival, but no mortality trial can establish that the observational estimate is causal. Evidence for cancer and dementia is dominated by cohorts; the largest dedicated cognitive trial was null. Small depression trials are favourable but heterogeneous, and fatty-liver benefits appear comparable to those of low-fat diets unless calorie restriction, activity, or polyphenol enrichment is added. The pattern changes inflammation, lipids, metabolites, and gut microbial functions, yet these intermediate responses cannot substitute for clinical outcomes. Overall, the Mediterranean diet is best understood as a supported cardiometabolic programme whose effectiveness depends on adherence and implementation, not as a collection of therapeutic foods or a universal preventive treatment.

Introduction

The Mediterranean diet occupies an unusual position in nutrition research because several of its major claims have been tested against clinical endpoints rather than only weight, cholesterol, or self-reported disease. Its evidence base includes randomized primary and secondary cardiovascular prevention, long prospective follow-up in multiple countries, and mechanistic trials that monitor adherence through biomarkers and food provision. At the same time, the label covers variable interventions, and much of the literature on mortality, cancer, dementia, and mental health remains observational. The central question is therefore not whether the pattern is “healthy” in the abstract, but which outcomes it changes, by how much, and under which comparison conditions.

The distinction between a diet and a supported programme is consequential. Endpoint trials supplied foods, dietitian contact, group sessions, and repeated monitoring, while newer studies add energy restriction, physical activity, or behaviour-

al support. Reviews of dietary patterns also find far more cohort studies than randomized mortality evidence, making healthy-user confounding a persistent alternative explanation for many broad associations. English et al. 2021 A rigorous interpretation must give randomized events more weight than biological plausibility, and prospective associations more weight than cross-sectional correlations, without treating any one trial as infallible.

Definition and measurement

The intervention is a family of related eating patterns rather than a prescribed menu. Reviews consistently identify a plant-rich core—vegetables, fruit, legumes, nuts, minimally refined cereals—combined with olive oil as the main added fat, regular fish, and lower consumption of red or processed meat and sweets. Dairy, eggs, alcohol, portion sizes, and frequencies vary across historical and modern definitions. Davis et al. 2015, Bach-Faig et al. 2011 The Seven Countries Study helped formulate the hypothesis through long-term contrasts among populations, but its ecolog-

ical and cohort observations were not randomized evidence. [Menotti & Puddu 2015](#)

Most studies convert the pattern into an adherence score. Some scores award points relative to sex-specific or study-specific medians, while others use fixed intake targets; trials often use the 14-item Mediterranean Diet Adherence Screener. [Bach et al. 2006](#), [Schröder et al. 2011](#) These tools make a whole-diet exposure tractable, but they are not interchangeable: a two-point difference can represent different food substitutions in different datasets, and scoring systems developed in Mediterranean populations may classify non-Mediterranean diets differently. [Hutchins-Wiese et al. 2022](#) Direct comparison within the MCC-Spain study found limited agreement among adherence indexes, showing that instrument choice can change who is labelled highly adherent. [Olmedo-Requena et al. 2019](#)

Synthesis quality compounds this exposure problem. An appraisal of 24 cardiovascular reviews found that complete satisfaction of adapted methodological criteria ranged from 8% to 75%; many reports omitted full search details, participant characteristics, or appropriate pooling methods. [Huedo-Medina et al. 2016](#) The implication is not that meta-analysis is uninformative, but that the label “systematic review” does not by itself resolve variation in diet definition, comparator, intervention support, or outcome measurement.

Cardiovascular endpoint trials

PREDIMED provides the largest randomized primary-prevention dataset, but only the corrected republication can be treated as evidence. A statistical investigation of 5,087 randomized trials detected improbable baseline distributions across journals, contributing to scrutiny of prominent reports. [Carlisle 2017](#) PREDIMED investigators

subsequently identified household members enrolled without individual randomization and two sites where allocation procedures departed from protocol. The 2013 report was retracted; the 2018 republication analysed the data without assuming fully individual randomization and added exclusions and cluster-robust sensitivity analyses. [Estruch et al. 2018](#)

In the corrected analysis, 7,447 adults at high cardiovascular risk were followed for a median of 4.8 years. Compared with low-fat advice, the primary composite of myocardial infarction, stroke, or cardiovascular death had a hazard ratio of 0.69 (95% confidence interval 0.53–0.91) for the extra-virgin-olive-oil arm and 0.72 (0.54–0.95) for the nut arm. Excluding 1,588 participants with known or suspected allocation departures yielded essentially the same estimates. [Estruch et al. 2018](#) The correction preserves a clinically important result while reducing the certainty that would attach to an entirely conventional individual randomized controlled trial.

Two secondary-prevention trials extend the finding beyond one study. The Lyon Diet Heart Study randomized 605 people after myocardial infarction and reported cardiac death or nonfatal infarction in 14 Mediterranean-pattern participants and 44 controls over 46 months. [de Lorgeril et al. 1999](#) Its effect was strikingly large and has not been reproduced at the same magnitude. CORDIOPREV randomized 1,002 patients with established coronary disease to Mediterranean or low-fat diets; after seven years, 87 versus 111 participants experienced a major event, corresponding to adjusted hazard ratios around 0.72–0.75. [Delgado-Lista et al. 2022](#) The three settings therefore agree on direction, while trial size, intervention composition, and effect magnitude differ.

TRIAL	SETTING	COMPARISON	MAIN ENDPOINT RESULT
PREDIMED republication	High-risk primary prevention; n=7,447	Olive oil or nuts vs low-fat advice	HR 0.69 and 0.72 for major cardiovascular events Estruch et al. 2018
Lyon Diet Heart	Post-infarction; n=605	Mediterranean pattern vs prudent Western diet	Cardiac death/nonfatal infarction: 14 vs 44 de Lorgeril et al. 1999
CORDIOPREV	Coronary disease; n=1,002	Mediterranean vs low-fat diet	87 vs 111 major events; adjusted HR about 0.72–0.75 Delgado-Lista et al. 2022

Comparative cardiovascular evidence

The most informative modern synthesis compares supported programmes with both minimal

advice and other active diets. A network meta-analysis of 40 trials and 35,548 higher-risk participants estimated that Mediterranean programmes reduced all-cause mortality (odds ratio

0.72), cardiovascular mortality (0.55), stroke (0.65), and nonfatal infarction (0.48) versus minimal intervention. At intermediate baseline risk, the estimates corresponded to about 17 fewer deaths, 13 fewer cardiovascular deaths, seven fewer strokes, and 17 fewer infarctions per 1,000 people over five years. [Karam et al. 2023](#) However, low-fat programmes also reduced mortality and infarction, and there was no convincing advantage of Mediterranean over low-fat programmes for those endpoints. The result supports structured dietary care more strongly than dietary exceptionalism.

Earlier syntheses were more cautious. Six randomized trials with 10,950 participants produced a risk ratio of 0.63 for major vascular events but 1.00 for all-cause mortality; excluding one trial with serious integrity concerns erased several favourable secondary findings. [Liyanage et al. 2016](#) A Cochrane review included 30 trials and 12,461 participants and emphasized low-to-moderate certainty, heterogeneous definitions, sparse events, and weak adverse-event reporting. [Rees et al. 2019](#) Another systematic review concluded that unrestricted-fat Mediterranean patterns improved several health outcomes, while noting that the clinical-event evidence depended heavily on a small number of trials. [Bloomfield et al. 2016](#)

Newer reviews continue to find fewer cardiovascular events without removing the structural dependence on PREDIMED. [Sebastian et al. 2024](#) The 2026 Italian-guideline syntheses included 87 primary-prevention studies with more than 1.4 million participants and 19 secondary-prevention studies with more than 91,000; they reported lower incident and recurrent cardiovascular risk, but combined randomized and cohort estimates and graded certainty separately by outcome. [Volpe et al. 2026a](#), [Volpe et al. 2026b](#) Risk-factor comparisons are smaller than event comparisons: Mediterranean diets modestly improve weight, pressure, glucose, and lipids relative to low-fat diets, while active-diet differences are narrower than contrasts with usual care. [Nordmann et al. 2011](#)

Mortality and longevity

Prospective cohorts show a remarkably stable dose-response association between Mediterranean adherence and survival. In 22,043 Greek adults, each two-point increase in the orig-

inal score corresponded to a 25% lower mortality rate; in 74,607 older participants across Europe, the analogous estimate was 8%. [Trichopoulou et al. 2003](#), [Trichopoulou et al. 2005](#) The difference between those estimates is informative: the association survives geographic expansion but becomes more modest.

Early pooling of 12 cohorts and 1.57 million participants estimated all-cause mortality risk ratio 0.91 per two score points. [Sofi et al. 2008](#) Subsequent syntheses remained close to a 9%–10% reduction per two points despite differences in score and follow-up. [Eleftheriou et al. 2018](#), [Soltani et al. 2019](#) The most recent broad meta-analysis screened 5,229 records and included 54 cohorts, 1.83 million participants, and 346,034 deaths. Its pooled risk ratio was 0.96 (95% CI 0.95–0.97) per one-point adherence increase, with moderate certainty. [Nucci et al. 2026](#) This is a graded association, not evidence that a single menu reduces mortality by a fixed percentage.

Large US cohorts support transfer beyond the Mediterranean. In the Nurses' Health Study and Health Professionals Follow-up Study, the top healthy-pattern categories were associated with total mortality HR 0.82 and cardiovascular disease HR 0.83. [Shan et al. 2023](#), [Shan et al. 2020](#) Among 25,315 US women followed for up to 25 years, top adherence corresponded to mortality HR 0.77; metabolite, inflammatory, triglyceride-rich lipoprotein, insulin-resistance, and body-mass pathways explained only part of the estimate. [Ahmad et al. 2024](#) People with diabetes also show an inverse association, with mortality HR 0.63 per two score points in the Moli-sani cohort. [Bonaccio et al. 2016](#)

Objective exposure measures help but do not convert cohorts into trials. In 642 older Italians with 20-year follow-up, a biomarker-based score predicted all-cause mortality HR 0.72 and cardiovascular mortality HR 0.55 for the highest versus lowest tertile, whereas repeated food-questionnaire scoring was weaker. Agreement between biomarker and questionnaire scores was low, biomarkers were measured only at baseline, and residual confounding remained possible. [Hidalgo-Liberona et al. 2021](#) The mortality literature is therefore consistent and dose-responsive, but its causal magnitude remains uncertain because no trial was powered to reproduce the cohort endpoint.

Diabetes, weight, and metabolic health

Type 2 diabetes is the non-cardiovascular outcome for which experimental, prospective, and biomarker evidence align most closely. In the diabetes-free PREDIMED subgroup, both Mediterranean interventions lowered incidence relative to low-fat advice. [Salas-Salvadó et al. 2014](#) An analysis using up to eight annual adherence measurements found HR 0.80 (95% CI 0.70–0.92) per two MEDAS points and 0.46 (0.25–0.83) for the highest versus lowest adherence category. [Martínez-González et al. 2023](#) Because attained adherence is partly behavioural rather than purely randomized, these estimates are strongest as evidence of a gradient within a trial context.

PREDIMED-Plus tested whether an intensive weight-loss programme added benefit. Among 4,746 adults free of diabetes, energy-restricted Mediterranean diet plus activity reduced six-year incidence by 31% relative to an unrestricted Mediterranean diet: 9.5% versus 12.0%. [Ruiz-Canela et al. 2025](#) This comparison is clinically useful but cannot isolate diet, energy deficit, exercise, and behavioural contact. In a separate 1,521-person imaging substudy, the same programme reduced total and visceral fat and attenuated relative lean-mass decline over three years, with partial weight regain after the first year. [Konieczna et al. 2023](#)

Objective biomarkers reinforce the diabetes association. A score based on five carotenoids and 24 fatty acids discriminated the Mediterranean arm of a controlled feeding trial with a C-statistic of 0.88, then predicted incident diabetes HR 0.71 per standard deviation in 22,202 EPIC-InterAct participants, versus HR 0.90 for self-report. [Sobiecki et al. 2023](#) The score may capture foods outside the Mediterranean pattern and still permits residual confounding, but it makes reporting bias alone an unlikely explanation. A broader Mediterranean-lifestyle index also predicted diabetes HR 0.70 for the highest versus lowest quartile in 112,493 UK Biobank adults, although the exposure included activity, rest, and social habits. [Maroto-Rodriguez et al. 2023](#)

Treatment effects in established diabetes are smaller. Prospective and trial reviews estimate lower incidence with adherence and modest improvements in glycaemic control relative to comparator diets. [Schwingshackl et al. 2015](#), [Esposito et al. 2015](#) A 2025 meta-analysis of 11 trials

found glycated haemoglobin lower by 0.31 percentage points, fasting glucose by 0.85 mmol/L, and body mass index by 0.83 kg/m². [Wu et al. 2025](#) These changes are useful adjuncts, not evidence that the diet replaces medication.

Weight-loss comparisons likewise argue against uniqueness. DIRECT randomized 322 adults and found two-year losses of 4.4 kg with Mediterranean, 2.9 kg with low-fat, and 4.7 kg with low-carbohydrate diets. [Shai et al. 2008](#) Across 121 randomized trials, most named diets produced modest six-month loss that diminished by 12 months; Mediterranean programmes were notable for retaining some cardiovascular risk-factor improvement, not for superior long-term weight loss. [Ge et al. 2020](#) A 50-study synthesis and a 57-trial review found small improvements across metabolic-syndrome components, with considerable heterogeneity. [Kastorini et al. 2011](#), [Papadaki et al. 2020](#) A green Mediterranean variant more than doubled visceral-fat loss relative to standard Mediterranean diet, but added walnuts, green tea, Mankai, energy restriction, and activity in a workforce that was 88% male. [Zelicha et al. 2022](#)

The null evidence identifies the appropriate unit of intervention. A 2024 meta-analysis of olive-oil-enriched Mediterranean trials found no consistent cardiometabolic advantage except a very small triglyceride reduction, so adding oil does not reproduce the whole pattern. [Keshani et al. 2024](#) In people with pre-existing metabolic disease, a 69-study synthesis found modest improvement in body mass index, waist, LDL cholesterol, triglycerides, fasting glucose, insulin resistance, and C-reactive protein, while HbA1c and lean-mass results were inconsistent. [Muscogiuri et al. 2026](#)

Cancer

Cancer is where breadth of association most clearly exceeds causal depth. An umbrella review judged evidence comparatively robust for overall cancer incidence and mortality but only suggestive or weak for many individual sites. [Dinu et al. 2018](#) The largest updated synthesis included 117 studies and 3.20 million participants. High adherence was associated with cancer mortality risk ratio 0.87, colorectal cancer 0.83, breast cancer 0.94, and larger reductions for some less common sites, yet most site-specific comparisons

were rated low or very low certainty. [Morze et al. 2021](#)

Repeated meta-analyses find the same broad direction but share the same observational inputs, scoring heterogeneity, and residual lifestyle confounding. [Schwingshackl et al. 2017](#) Dose-response analysis supports lower colorectal risk, and a cohort/meta-analysis of postmenopausal breast cancer found the inverse association concentrated in estrogen-receptor-negative tumours. [Zhong et al. 2020](#), [van den Brandt & Schulpen 2017](#) These subgroup patterns can be biologically interesting without being intervention effects.

Randomized evidence consists mainly of one small secondary outcome. In PREDIMED, the olive-oil group experienced fewer invasive breast cancers than low-fat controls, HR 0.32 (95% CI 0.13–0.79), but only 35 cases occurred and cancer was not the trial's primary endpoint. [Toledo et al. 2015](#) A breast-cancer umbrella review reaches a favourable overall interpretation while acknowledging the predominance of observational evidence. [González-Palacios Torres et al. 2023](#) Olive-oil intake itself is associated with lower cardiovascular disease, diabetes, and mortality, but its pooled cancer estimate is null at risk ratio 0.94 (0.86–1.03). [Martínez-González et al. 2022](#) The cancer literature therefore supports prevention hypotheses and a generally plant-rich pattern, not a claim that Mediterranean diet treatment prevents specific cancers.

Cognition and dementia

Prospective studies associate Mediterranean adherence with lower cognitive decline. A 2025 meta-analysis of 23 studies reported HR 0.82 for cognitive impairment, 0.89 for dementia, and 0.70 for Alzheimer disease, while documenting heterogeneity, possible publication bias, and limited non-Western representation. [Fekete et al. 2025](#) A separate updated synthesis also found inverse dementia and Alzheimer associations. [Nucci et al. 2024](#) UK Biobank provides a useful absolute-risk estimate: among 60,298 adults, high adherence corresponded to dementia HR 0.77 and absolute risk 1.18% versus 1.73% over 9.1 years, without measurable interaction with polygenic risk. [Shannon et al. 2023](#)

Earlier reviews combined favourable cohorts with small trials. One synthesis found mild cognitive impairment risk ratio 0.75 and Alzheimer disease 0.71 in cohorts; two trials yielded an episodic-

memory standardized mean difference of 0.20. [Fu et al. 2022](#) Another meta-analysis reached a similar observational conclusion. [Singh et al. 2014](#) A PREDIMED-Navarra substudy reported better global cognition after 6.5 years, but it was small and inherited the allocation limitations of the parent trial. [Martínez-Lapiscina et al. 2013](#)

The largest dedicated trial sharply lowers expectations. The three-year MIND trial randomized 604 older adults with suboptimal diets and obesity history; both intervention and control were calorie-restricted. Global cognition differed by only 0.035 standard deviations (95% CI –0.022 to 0.092), and brain MRI outcomes did not differ. [Barnes et al. 2023](#) A MIND-diet meta-analysis remains favourable because its pooled estimate is dominated by cohorts, whereas a 2024 PREVENT analysis found no association with cognition or neuroimaging. [Chen et al. 2023](#), [Gregory et al. 2024](#) The most defensible synthesis is that long-term adherence may mark lower dementia risk, but a three-year supported MIND intervention did not improve cognition in relatively healthy older volunteers.

Depression and mental health

Prospective cohorts associate Mediterranean adherence with lower incident depression. Meta-analyses report an inverse association of roughly one-third for the highest adherence groups, although reverse causation is difficult to exclude because early depression can alter shopping, appetite, activity, and diet quality. [Lassale et al. 2019](#), [Shafiei et al. 2019](#) The observational result is therefore compatible with benefit without proving treatment efficacy.

Small randomized trials provide a causal signal but an unstable magnitude. SMILES randomized 67 adults with major depression to dietary support or befriending-style social support for 12 weeks. The diet group improved 7.1 points more on the [Montgomery-Åsberg Depression Rating Scale](#), and remission was 32% versus 8%. [Jacka et al. 2017](#) HELFIMED enrolled 152 adults and found greater symptom improvement with group Mediterranean education plus fish-oil supplementation than with social groups, a design that cannot isolate the diet. [Parletta et al. 2019](#) AMMEND extended the result to 72 young men over 12 weeks, again in a small, selected population. [Bayes et al. 2022](#)

Pooling magnifies the uncertainty. Five trials with 1,507 participants yielded symptom standardized mean difference -0.53 , but heterogeneity was 87%, the prediction interval crossed zero, and certainty was low. [Bizzozero-Peroni et al. 2025](#) A 2025 review identified 104 mental-health studies—88 observational and only 16 interventional—and found favourable associations with depression, anxiety, stress, quality of life, and well-being. [Kabthymmer et al. 2026](#) Direction is consistent enough to justify larger trials, but current evidence does not identify a stable antidepressant effect or support replacing established care.

Fatty liver

Mediterranean interventions reduce hepatic fat, but direct comparison weakens claims of unique superiority. In a 12-week randomized trial, ad libitum Mediterranean and low-fat diets both reduced steatosis without a decisive between-diet difference. [Properzi et al. 2018](#) In DIRECT, decreased hepatic fat statistically mediated part of the Mediterranean diet's cardiometabolic advantage over low fat, but mediation cannot identify the responsible foods. [Gepner et al. 2019](#)

The largest imaging effects come from an enriched, energy-restricted variant. DIRECT-PLUS randomized 294 adults to healthy guidance, standard Mediterranean, or green Mediterranean diets, with physical activity in every arm. Relative liver-fat reduction was 12.2%, 19.6%, and 38.9%, respectively, after 18 months; the green arm added walnuts, green tea, Mankai, and further red-meat restriction. [Yaskolka Meir et al. 2021](#) This shows that a complex enriched programme can change liver fat, not that olive oil alone treats liver disease.

Meta-analyses explain the mixed interpretation. Earlier randomized-trial pooling found improvement in steatosis and insulin resistance and small, inconsistent enzyme changes. [Kawaguchi et al. 2021](#), [Sangouni et al. 2022](#) A 2024 head-to-head review concluded that Mediterranean and low-fat diets had similar short-term effects on liver enzymes and fat content. [Xiong et al. 2024](#) An Asian-adapted randomized trial suggests the approach can be transferred culturally, but adaptation changes the tested foods and long-term clinical liver outcomes remain unavailable. [Chooi et al. 2024](#)

Biological pathways

Mechanistic evidence supports plausibility through several pathways rather than one “active ingredient.” Trial syntheses report small changes in endothelial function and selected inflammatory markers, with high heterogeneity across assays and populations. [Schwingshackl & Hoffmann 2014](#), [Koelman et al. 2022](#) A 2026 review of 33 randomized trials and 3,476 participants found reductions in high-sensitivity C-reactive protein, interleukin-6, and interleukin-17, but not conventional CRP, interleukin-10, tumour-necrosis factor, or total antioxidant capacity. [Keshani et al. 2026](#) Reviews therefore implicate fat quality, plant polyphenols, fibre fermentation, inflammation, endothelial function, and insulin sensitivity without establishing one required component. [Itsipoulos et al. 2022](#)

Controlled feeding and isocaloric designs separate dietary composition from weight loss. In an eight-week trial of 82 overweight or obese adults, a weight-maintaining Mediterranean diet lowered total cholesterol and altered faecal bile acids, microbial functions, and metabolites. Glucose, insulin, and CRP did not change. [Meslier et al. 2020](#) In older adults, the one-year NU-AGE intervention shifted microbial taxa and functions that covaried with lower frailty and inflammation; the analysed microbiome subset included 612 participants across five countries. [Ghosh et al. 2020](#) A separate obesity trial found that both Mediterranean and low-fat healthy diets modified gut communities alongside improved insulin sensitivity. [Haro et al. 2016](#)

Molecular profiles are also becoming exposure measures. Plasma metabolomic patterns tracked Mediterranean adherence and subsequent cardiovascular risk, though they cannot by themselves prove mediation. [Li et al. 2020](#) In DIRECT-PLUS, baseline bile acids and particular microbial species modified lipid and adiposity responses, while a PREDIMED-Plus analysis connected intervention-related microbiome and metabolome shifts with risk-factor changes. [Gao et al. 2024](#), [García-Gavilán et al. 2024](#) These studies clarify how people may respond differently; they do not upgrade surrogate changes into evidence for fewer cancers, dementia cases, or deaths.

Implementation, cost, and sustainability

Transfer outside Mediterranean settings requires adaptation rather than literal replication.

Reviews emphasize matching the pattern to local staples, availability, culinary practice, and cultural identity while preserving its plant-forward structure and fat quality. [Hoffman & Gerber 2013](#), [Woodside et al. 2022](#) A Mediterranean score developed around olive oil, wine, and regional medians may misclassify an equally plant-rich pattern elsewhere. Implementation is therefore a question of function—what foods replace refined grains, processed meat, and saturated fat—not geographic authenticity.

Supported efficacy also exceeds unsupported adoption. MedEx-UK showed that a theory-informed programme could improve adherence and was acceptable to older adults at dementia risk, but it used behavioural maintenance and physical-activity components. [Jennings et al. 2024](#) Cross-country studies identify cost, access, cooking skills, family norms, and nutritional knowledge as barriers. [Biggi et al. 2024](#) PREDIMED supplied oil or nuts; PREDIMED-Plus added frequent dietitian contact and activity coaching. Real-world benefit will decline if those adherence supports disappear.

Economic and environmental effects depend on assumptions. A secondary-prevention analysis es-

timated Mediterranean diet plus activity to be cost-effective, but the model inherits adherence and event-reduction estimates from other data. [Bonekamp et al. 2024](#) A systematic review of 35 sustainability studies—25 modelling, seven cross-sectional, and three longitudinal—reported the pattern environmentally favourable in 91% of comparisons. Most gains came from lower ruminant-meat intake, while water demand and local production could change particular rankings. [Lorca-Camara et al. 2024](#) The Med Diet 4.0 framework similarly treats health, environment, culture, and local economy as linked dimensions rather than interchangeable outcomes. [Dernini et al. 2017](#)

Alcohol requires a modern qualification. Traditional scores often award a point for moderate wine, but alcohol raises cancer, dependence, and injury risk. A contemporary review argues that wine is not necessary to preserve the pattern’s benefits and should not be recommended to non-drinkers. [Martínez-González 2024](#) The therapeutic interpretation is therefore vegetables, legumes, whole grains, nuts, fish, and unsaturated fats—not an instruction to begin drinking.

Overall certainty

OUTCOME	CURRENT CONCLUSION	MAIN REASON CONFIDENCE IS LIMITED
Major cardiovascular events	Probably reduced in higher-risk adults, particularly versus minimal support	Few endpoint trials; corrected PREDIMED carries much of primary-prevention weight
Type 2 diabetes	Probably prevented; established disease improves modestly	Intensive programmes combine diet, energy restriction, activity, and support
All-cause mortality	Consistent 4% lower risk per adherence point in cohorts	No randomized mortality confirmation; residual healthy-user bias
Weight and risk factors	Small favourable changes	Similar benefits occur with other supported healthy diets
Cancer	Several inverse site-specific associations	Predominantly observational; one small randomized secondary signal
Cognition	Cohorts favourable, largest dedicated trial null	Active control, limited duration, exposure confounding
Depression	Small trials favourable	Low certainty, high heterogeneity, weak blinding
Fatty liver	Liver fat decreases	Short-term equivalence to low-fat diets; few clinical outcomes

The evidence hierarchy is consequently uneven. Cardiovascular events and diabetes prevention combine randomized and observational support. Mortality has scale, dose-response, and geographic replication but not randomized confirmation. Cancer and dementia are mostly cohort findings, depression is a small-trial signal, and liver studies mainly measure imaging or en-

zymes. Mechanistic responses make the cardiometabolic findings coherent, but they cannot fill causal gaps for unrelated outcomes.

Conclusion

The Mediterranean diet has a defensible, bounded set of benefits. Structured programmes probably reduce major cardiovascular events in adults

at increased risk, especially compared with minimal advice; they are not clearly superior to every supported low-fat programme. Type 2 diabetes is probably prevented, and established diabetes, body composition, and metabolic risk factors improve by modest amounts. Higher adherence is consistently associated with longer survival, but the size of that effect remains observational.

Claims should be weaker for other outcomes. Cancer associations cover several sites without adequate prevention trials. Dementia cohorts are favourable, yet the largest dedicated cognitive trial was null. Depression trials point toward benefit but are too small and heterogeneous to define a treatment effect. Energy-restricted and enriched variants reduce liver fat, while standard Mediterranean and low-fat diets perform similarly in direct short-term comparison. The practical conclusion is not that olive oil, nuts, wine, or polyphenols are medicines. It is that replacing processed meat, refined foods, and saturated fat with a supported, culturally workable plant-forward pattern has credible cardiometabolic value, with the remaining claims still awaiting stronger trials.

REFERENCES

- Ahmad et al. 2024** Mediterranean Diet Adherence and Risk of All-Cause Mortality in Women. *JAMA Network Open*. <https://doi.org/10.1001/jamanetworkopen.2024.14322>
- Bach et al. 2006** The use of indexes evaluating the adherence to the Mediterranean diet in epidemiological studies: a review. *Public Health Nutrition*. <https://doi.org/10.1079/phn2005936>
- Bach-Faig et al. 2011** Mediterranean diet pyramid today. Science and cultural updates. *Public Health Nutrition*. <https://doi.org/10.1017/s1368980011002515>
- Barnes et al. 2023** Trial of the MIND Diet for Prevention of Cognitive Decline in Older Persons. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2302368>
- Bayes et al. 2022** The effect of a Mediterranean diet on the symptoms of depression in young males (the “AM-MEND: A Mediterranean Diet in MEN with Depression” study): a randomized controlled trial. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1093/ajcn/nqac106>
- Biggi et al. 2024** Drivers and Barriers Influencing Adherence to the Mediterranean Diet: A Comparative Study across Five Countries. *Nutrients*. <https://doi.org/10.3390/nu16152405>
- Bizzozero-Peroni et al. 2025** The impact of the Mediterranean diet on alleviating depressive symptoms in adults: a systematic review and meta-analysis of randomized controlled trials. *Nutrition Reviews*. <https://doi.org/10.1093/nutrit/nuad176>
- Bloomfield et al. 2016** Effects on Health Outcomes of a Mediterranean Diet With No Restriction on Fat Intake. *Annals of Internal Medicine*. <https://doi.org/10.7326/m16-0361>
- Bonaccio et al. 2016** Adherence to the traditional Mediterranean diet and mortality in subjects with diabetes. Prospective results from the MOLI-SANI study. *European Journal of Preventive Cardiology*. <https://doi.org/10.1177/2047487315569409>
- Bonekamp et al. 2024** Cost-effectiveness of Mediterranean diet and physical activity in secondary cardiovascular disease prevention: results from the UCC-SMART cohort study. *European Journal of Preventive Cardiology*. <https://doi.org/10.1093/eurjpc/zwae123>
- Carlisle 2017** Data fabrication and other reasons for non-random sampling in 5087 randomised, controlled trials in anaesthetic and general medical journals. *Anaesthesia*. <https://doi.org/10.1111/anae.13938>
- Chen et al. 2023** Association of the Mediterranean Dietary Approaches to Stop Hypertension Intervention for Neurodegenerative Delay (MIND) Diet With the Risk of Dementia. *JAMA Psychiatry*. <https://doi.org/10.1001/jamapsychiatry.2023.0800>
- Chooi et al. 2024** Effect of an Asian-adapted Mediterranean diet and pentadecanoic acid on fatty liver disease: the TANGO randomized controlled trial. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1016/j.ajcnut.2023.11.013>
- Davis et al. 2015** Definition of the Mediterranean Diet; A Literature Review. *Nutrients*. <https://doi.org/10.3390/nu7115459>
- de Lorgeril et al. 1999** Mediterranean Diet, Traditional Risk Factors, and the Rate of Cardiovascular Complications After Myocardial Infarction. *Circulation*. <https://doi.org/10.1161/01.cir.99.6.779>
- Delgado-Lista et al. 2022** Long-term secondary prevention of cardiovascular disease with a Mediterranean diet and a low-fat diet (CORDIOPREV): a randomised controlled trial. *The Lancet*. [https://doi.org/10.1016/s0140-6736\(22\)00122-2](https://doi.org/10.1016/s0140-6736(22)00122-2)
- Dernini et al. 2017** Med Diet 4.0: the Mediterranean diet with four sustainable benefits. *Public Health Nutrition*. <https://doi.org/10.1017/s1368980016003177>
- Dinu et al. 2018** Mediterranean diet and multiple health outcomes: an umbrella review of meta-analyses of observational studies and randomised trials. *European Journal of Clinical Nutrition*. <https://doi.org/10.1038/ejcn.2017.58>
- Eleftheriou et al. 2018** Mediterranean diet and its components in relation to all-cause mortality: meta-analysis. *British Journal of Nutrition*. <https://doi.org/10.1017/s0007114518002593>
- English et al. 2021** Evaluation of Dietary Patterns and All-Cause Mortality. *JAMA Network Open*. <https://doi.org/10.1001/jamanetworkopen.2021.22277>
- Esposito et al. 2015** A journey into a Mediterranean diet and type 2 diabetes: a systematic review with meta-

- analyses. *BMJ Open*. <https://doi.org/10.1136/bmjopen-2015-008222>
- Estruch et al. 2018** Primary Prevention of Cardiovascular Disease with a Mediterranean Diet Supplemented with Extra-Virgin Olive Oil or Nuts. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa1800389>
- Fekete et al. 2025** The role of the Mediterranean diet in reducing the risk of cognitive impairment, dementia, and Alzheimer's disease: a meta-analysis. *GeroScience*. <https://doi.org/10.1007/s11357-024-01488-3>
- Fu et al. 2022** Association between the mediterranean diet and cognitive health among healthy adults: A systematic review and meta-analysis. *Frontiers in Nutrition*. <https://doi.org/10.3389/fnut.2022.946361>
- Gao et al. 2024** Gut microbial metabolism of bile acids modifies the effect of Mediterranean diet interventions on cardiometabolic risk in a randomized controlled trial. *Gut Microbes*. <https://doi.org/10.1080/19490976.2024.2426610>
- García-Gavilán et al. 2024** Effect of 1-year lifestyle intervention with energy-reduced Mediterranean diet and physical activity promotion on the gut metabolome and microbiota: a randomized clinical trial. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1016/j.ajcnut.2024.02.021>
- Ge et al. 2020** Comparison of dietary macronutrient patterns of 14 popular named dietary programmes for weight and cardiovascular risk factor reduction in adults: systematic review and network meta-analysis of randomised trials. *BMJ*. <https://doi.org/10.1136/bmj.m696>
- Gepner et al. 2019** The beneficial effects of Mediterranean diet over low-fat diet may be mediated by decreasing hepatic fat content. *Journal of Hepatology*. <https://doi.org/10.1016/j.jhep.2019.04.013>
- Ghosh et al. 2020** Mediterranean diet intervention alters the gut microbiome in older people reducing frailty and improving health status: the NU-AGE 1-year dietary intervention across five European countries. *Gut*. <https://doi.org/10.1136/gutjnl-2019-319654>
- González-Palacios Torres et al. 2023** Mediterranean diet and risk of breast cancer: An umbrella review. *Clinical Nutrition*. <https://doi.org/10.1016/j.clnu.2023.02.012>
- Gregory et al. 2024** The Mediterranean diet is not associated with neuroimaging or cognition in middle-aged adults: a cross-sectional analysis of the PREVENT dementia programme. *European Journal of Neurology*. <https://doi.org/10.1111/ene.16345>
- Haro et al. 2016** Two Healthy Diets Modulate Gut Microbial Community Improving Insulin Sensitivity in a Human Obese Population. *The Journal of Clinical Endocrinology & Metabolism*. <https://doi.org/10.1210/aj.2015-3351>
- Hidalgo-Liberona et al. 2021** Adherence to the Mediterranean diet assessed by a novel dietary biomarker score and mortality in older adults: the InCHIANTI cohort study. *BMC Medicine*. <https://doi.org/10.1186/s12916-021-02154-7>
- Hoffman & Gerber 2013** Evaluating and adapting the Mediterranean diet for non-Mediterranean populations: A critical appraisal. *Nutrition Reviews*. <https://doi.org/10.1111/nure.12040>
- Huedo-Medina et al. 2016** Methodologic quality of meta-analyses and systematic reviews on the Mediterranean diet and cardiovascular disease outcomes: a review. *The American Journal of Clinical Nutrition*. <https://doi.org/10.3945/ajcn.115.112771>
- Hutchins-Wiese et al. 2022** Mediterranean diet scoring systems: understanding the evolution and applications for Mediterranean and non-Mediterranean countries. *British Journal of Nutrition*. <https://doi.org/10.1017/s0007114521002476>
- Itsiopoulos et al. 2022** The anti-inflammatory effects of a Mediterranean diet: a review. *Current Opinion in Clinical Nutrition & Metabolic Care*. <https://doi.org/10.1097/mco.0000000000000872>
- Jacka et al. 2017** A randomised controlled trial of dietary improvement for adults with major depression (the 'SMILES' trial). *BMC Medicine*. <https://doi.org/10.1186/s12916-017-0791-y>
- Jennings et al. 2024** Effectiveness and feasibility of a theory-informed intervention to improve Mediterranean diet adherence, physical activity and cognition in older adults at risk of dementia: the MedEx-UK randomised controlled trial. *BMC Medicine*. <https://doi.org/10.1186/s12916-024-03815-z>
- Kabthymmer et al. 2026** Effect of Mediterranean diet on mental health outcomes: a systematic review. *Nutrition Research Reviews*. <https://doi.org/10.1017/s0954422425100243>
- Karam et al. 2023** Comparison of seven popular structured dietary programmes and risk of mortality and major cardiovascular events in patients at increased cardiovascular risk: systematic review and network meta-analysis. *BMJ*. <https://doi.org/10.1136/bmj-2022-072003>
- Kastorini et al. 2011** The Effect of Mediterranean Diet on Metabolic Syndrome and its Components. *Journal of the American College of Cardiology*. <https://doi.org/10.1016/j.jacc.2010.09.073>
- Kawaguchi et al. 2021** Effects of Mediterranean Diet in Patients with Nonalcoholic Fatty Liver Disease: A Systematic Review, Meta-Analysis, and Meta-Regression Analysis of Randomized Controlled Trials. *Seminars in Liver Disease*. <https://doi.org/10.1055/s-0041-1723751>
- Keshani et al. 2024** Mediterranean diet enriched with olive oil shows no consistent benefits on cardiometabolic and anthropometric parameters: a systematic review with meta-analysis of randomized controlled trials. *European Journal of Nutrition*. <https://doi.org/10.1007/s00394-024-03478-9>
- Keshani et al. 2026** Mediterranean Diet Reduces Inflammation in Adults: A Systematic Review and Meta-analysis of Randomized Controlled Trials. *Nutrition Reviews*. <https://doi.org/10.1093/nutrit/nuaf213>
- Koelman et al. 2022** Effects of Dietary Patterns on Biomarkers of Inflammation and Immune Responses: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Advances in Nutrition*. <https://doi.org/10.1093/advances/nmab086>

- Konieczna et al. 2023** An Energy-Reduced Mediterranean Diet, Physical Activity, and Body Composition. *JAMA Network Open*. <https://doi.org/10.1001/jamanetworkopen.2023.37994>
- Lassale et al. 2019** Healthy dietary indices and risk of depressive outcomes: a systematic review and meta-analysis of observational studies. *Molecular Psychiatry*. <https://doi.org/10.1038/s41380-018-0237-8>
- Li et al. 2020** The Mediterranean diet, plasma metabolome, and cardiovascular disease risk. *European Heart Journal*. <https://doi.org/10.1093/eurheartj/ehaa209>
- Liyanage et al. 2016** Effects of the Mediterranean Diet on Cardiovascular Outcomes—A Systematic Review and Meta-Analysis. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0159252>
- Lorca-Camara et al. 2024** Environmental and Health Sustainability of the Mediterranean Diet: A Systematic Review. *Advances in Nutrition*. <https://doi.org/10.1016/j.advnut.2024.100322>
- Maroto-Rodríguez et al. 2023** Association between a mediterranean lifestyle and Type 2 diabetes incidence: a prospective UK biobank study. *Cardiovascular Diabetology*. <https://doi.org/10.1186/s12933-023-01999-x>
- Martínez-González et al. 2022** Effect of olive oil consumption on cardiovascular disease, cancer, type 2 diabetes, and all-cause mortality: A systematic review and meta-analysis. *Clinical Nutrition*. <https://doi.org/10.1016/j.clnu.2022.10.001>
- Martínez-González et al. 2023** Yearly attained adherence to Mediterranean diet and incidence of diabetes in a large randomized trial. *Cardiovascular Diabetology*. <https://doi.org/10.1186/s12933-023-01994-2>
- Martínez-González 2024** Should we remove wine from the Mediterranean diet?: a narrative review. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1016/j.ajcnut.2023.12.020>
- Martínez-Lapiscina et al. 2013** Mediterranean diet improves cognition: the PREDIMED-NAVARRA randomised trial. *Journal of Neurology, Neurosurgery & Psychiatry*. <https://doi.org/10.1136/jnnp-2012-304792>
- Menotti & Puddu 2015** How the Seven Countries Study contributed to the definition and development of the Mediterranean diet concept: A 50-year journey. *Nutrition, Metabolism and Cardiovascular Diseases*. <https://doi.org/10.1016/j.numecd.2014.12.001>
- Meslier et al. 2020** Mediterranean diet intervention in overweight and obese subjects lowers plasma cholesterol and causes changes in the gut microbiome and metabolome independently of energy intake. *Gut*. <https://doi.org/10.1136/gutjnl-2019-320438>
- Morze et al. 2021** An updated systematic review and meta-analysis on adherence to mediterranean diet and risk of cancer. *European Journal of Nutrition*. <https://doi.org/10.1007/s00394-020-02346-6>
- Muscogiuri et al. 2026** Mediterranean diet for the management of pre-existing metabolic diseases: Evidence from a systematic review and meta-analysis featured in the Italian national guidelines “La Dieta Mediterranea”. *Nutrition*. <https://doi.org/10.1016/j.nut.2025.112975>
- Nordmann et al. 2011** Meta-Analysis Comparing Mediterranean to Low-Fat Diets for Modification of Cardiovascular Risk Factors. *The American Journal of Medicine*. <https://doi.org/10.1016/j.amjmed.2011.04.024>
- Nucci et al. 2024** Association between Mediterranean diet and dementia and Alzheimer disease: a systematic review with meta-analysis. *Aging Clinical and Experimental Research*. <https://doi.org/10.1007/s40520-024-02718-6>
- Nucci et al. 2026** Adherence to the Mediterranean diet and all-cause mortality: A systematic review and meta-analysis within the Italian National Guidelines “La Dieta Mediterranea”. *Nutrition*. <https://doi.org/10.1016/j.nut.2026.113189>
- Olmedo-Requena et al. 2019** Agreement among Mediterranean Diet Pattern Adherence Indexes: MCC-Spain Study. *Nutrients*. <https://doi.org/10.3390/nu11030488>
- Papadaki et al. 2020** The Effect of the Mediterranean Diet on Metabolic Health: A Systematic Review and Meta-Analysis of Controlled Trials in Adults. *Nutrients*. <https://doi.org/10.3390/nu12113342>
- Parletta et al. 2019** A Mediterranean-style dietary intervention supplemented with fish oil improves diet quality and mental health in people with depression: A randomized controlled trial (HELFIMED). *Nutritional Neuroscience*. <https://doi.org/10.1080/1028415x.2017.1411320>
- Properzi et al. 2018** Ad Libitum Mediterranean and Low-Fat Diets Both Significantly Reduce Hepatic Steatosis: A Randomized Controlled Trial. *Hepatology*. <https://doi.org/10.1002/hep.30076>
- Rees et al. 2019** Mediterranean-style diet for the primary and secondary prevention of cardiovascular disease. *Cochrane Database of Systematic Reviews*. <https://doi.org/10.1002/14651858.cd009825.pub3>
- Ruiz-Canela et al. 2025** Comparison of an Energy-Reduced Mediterranean Diet and Physical Activity Versus an Ad Libitum Mediterranean Diet in the Prevention of Type 2 Diabetes. *Annals of Internal Medicine*. <https://doi.org/10.7326/annals-25-00388>
- Salas-Salvadó et al. 2014** Prevention of Diabetes With Mediterranean Diets. *Annals of Internal Medicine*. <https://doi.org/10.7326/m13-1725>
- Sangouni et al. 2022** Effect of Mediterranean diet on liver enzymes: a systematic review and meta-analysis of randomised controlled trials. *British Journal of Nutrition*. <https://doi.org/10.1017/s0007114521002270>
- Schröder et al. 2011** A Short Screener Is Valid for Assessing Mediterranean Diet Adherence among Older Spanish Men and Women. *The Journal of Nutrition*. <https://doi.org/10.3945/jn.110.135566>
- Schwingshackl & Hoffmann 2014** Mediterranean dietary pattern, inflammation and endothelial function: A systematic review and meta-analysis of intervention trials. *Nutrition, Metabolism and Cardiovascular Diseases*. <https://doi.org/10.1016/j.numecd.2014.03.003>
- Schwingshackl et al. 2015** Adherence to a Mediterranean diet and risk of diabetes: a systematic review and meta-

- analysis. *Public Health Nutrition*. <https://doi.org/10.1017/s1368980014001542>
- Schwingshackl et al. 2017** Adherence to Mediterranean Diet and Risk of Cancer: An Updated Systematic Review and Meta-Analysis. *Nutrients*. <https://doi.org/10.3390/nu9101063>
- Sebastian et al. 2024** Long-term impact of mediterranean diet on cardiovascular disease prevention: A systematic review and meta-analysis of randomized controlled trials. *Current Problems in Cardiology*. <https://doi.org/10.1016/j.cpcardiol.2024.102509>
- Shafiei et al. 2019** Adherence to the Mediterranean diet and risk of depression: a systematic review and updated meta-analysis of observational studies. *Nutrition Reviews*. <https://doi.org/10.1093/nutrit/nuy070>
- Shai et al. 2008** Weight Loss with a Low-Carbohydrate, Mediterranean, or Low-Fat Diet. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa0708681>
- Shan et al. 2020** Association Between Healthy Eating Patterns and Risk of Cardiovascular Disease. *JAMA Internal Medicine*. <https://doi.org/10.1001/jamainternmed.2020.2176>
- Shan et al. 2023** Healthy Eating Patterns and Risk of Total and Cause-Specific Mortality. *JAMA Internal Medicine*. <https://doi.org/10.1001/jamainternmed.2022.6117>
- Shannon et al. 2023** Mediterranean diet adherence is associated with lower dementia risk, independent of genetic predisposition: findings from the UK Biobank prospective cohort study. *BMC Medicine*. <https://doi.org/10.1186/s12916-023-02772-3>
- Singh et al. 2014** Association of Mediterranean Diet with Mild Cognitive Impairment and Alzheimer's Disease: A Systematic Review and Meta-Analysis. *Journal of Alzheimer's Disease*. <https://doi.org/10.3233/jad-130830>
- Sobiecki et al. 2023** A nutritional biomarker score of the Mediterranean diet and incident type 2 diabetes: Integrated analysis of data from the MedLey randomised controlled trial and the EPIC-InterAct case-cohort study. *PLOS Medicine*. <https://doi.org/10.1371/journal.pmed.1004221>
- Sofi et al. 2008** Adherence to Mediterranean diet and health status: meta-analysis. *BMJ*. <https://doi.org/10.1136/bmj.a1344>
- Soltani et al. 2019** Adherence to the Mediterranean Diet in Relation to All-Cause Mortality: A Systematic Review and Dose-Response Meta-Analysis of Prospective Cohort Studies. *Advances in Nutrition*. <https://doi.org/10.1093/advances/nmz041>
- Toledo et al. 2015** Mediterranean Diet and Invasive Breast Cancer Risk Among Women at High Cardiovascular Risk in the PREDIMED Trial. *JAMA Internal Medicine*. <https://doi.org/10.1001/jamainternmed.2015.4838>
- Trichopoulou et al. 2003** Adherence to a Mediterranean Diet and Survival in a Greek Population. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa025039>
- Trichopoulou et al. 2005** Modified Mediterranean diet and survival: EPIC-elderly prospective cohort study. *BMJ*. <https://doi.org/10.1136/bmj.38415.644155.8f>
- van den Brandt & Schulpen 2017** Mediterranean diet adherence and risk of postmenopausal breast cancer: results of a cohort study and meta-analysis. *International Journal of Cancer*. <https://doi.org/10.1002/ijc.30654>
- Volpe et al. 2026a** Effectiveness of Mediterranean diet for the primary prevention of cardiovascular diseases: A systematic review and meta-analysis featured in the Italian National Guidelines “La Dieta Mediterranea”. *Nutrition*. <https://doi.org/10.1016/j.nut.2025.113038>
- Volpe et al. 2026b** Efficacy of Mediterranean diet for the prevention of cardiovascular disease in patients: A systematic review and meta-analysis featured in the Italian National Guidelines “La Dieta Mediterranea”. *Nutrition*. <https://doi.org/10.1016/j.nut.2025.113053>
- Woodside et al. 2022** Culturally adapting the Mediterranean Diet pattern – a way of promoting more ‘sustainable’ dietary change? *British Journal of Nutrition*. <https://doi.org/10.1017/s0007114522001945>
- Wu et al. 2025** Impact of the Mediterranean Diet on Glycemic Control, Body Mass Index, Lipid Profile, and Blood Pressure in Type 2 Diabetes: A Meta-Analysis of Randomized Controlled Trials. *Nutrients*. <https://doi.org/10.3390/nu17243908>
- Xiong et al. 2024** A systematic review and meta-analysis of randomized controlled trials: effects of mediterranean diet and low-fat diet on liver enzymes and liver fat content of NAFLD. *Food & Function*. <https://doi.org/10.1039/d4fo01461h>
- Yaskolka Meir et al. 2021** Effect of green-Mediterranean diet on intrahepatic fat: the DIRECT PLUS randomised controlled trial. *Gut*. <https://doi.org/10.1136/gutjnl-2020-323106>
- Zelicha et al. 2022** The effect of high-polyphenol Mediterranean diet on visceral adiposity: the DIRECT PLUS randomized controlled trial. *BMC Medicine*. <https://doi.org/10.1186/s12916-022-02525-8>
- Zhong et al. 2020** Association between Mediterranean diet adherence and colorectal cancer: a dose-response meta-analysis. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1093/ajcn/nqaa083>